

Discrete Structures

Problem Set 1

Release Date: August 26th, 2026

Deadline: September 3rd, 2026

Instructions:

- The marks for each question (or sub-question) has been specified alongside it.
- Take care to ensure that if you use variables that have not been defined in the question, you state them clearly, and the domain they belong to clearly as well(if applicable)
- Answer questions precisely, maintaining mathematical rigor as much as possible
- **Both** digital (**Written , Not Typed**) and handwritten submissions will be accepted.
- The deadline for this set is **September 3rd , 11:59 p.m.**
- Submissions to be done on moodle with the format:
<rollnumber>_DS_PS1 (for example 2023113019_DS_PS1)
- Plagiarism and GenAI use shall be penalized heavily.

Problem 1:

(2+2=4)

Consider the following pairs of propositions.

- (a) Determine, using a truth table, whether the following propositions are logically equivalent:

$$\neg(p \wedge q) \vee r \quad \text{and} \quad (\neg p \vee \neg q) \vee r.$$

- (b) Determine, using a truth table, whether the following propositions are logically equivalent:

$$(p \vee q) \wedge (p \vee \neg q) \quad \text{and} \quad p.$$

For each part, clearly state whether the two propositions are logically equivalent based on your truth table.

Problem 2:

(1+1+1+1=4)

Let the domain consist of all students enrolled at a university. Define the predicates

$C(x)$: “student x has completed a course in discrete mathematics,”

and

$P(x)$: “student x has completed a course in programming.”

Translate each of the following statements into a formal logical expression using the predicates $C(x)$ and $P(x)$, quantifiers ($\forall, \exists$), and logical connectives.

- (a) Every student who has completed discrete mathematics has also completed programming.

- (b) There is a student who has completed programming but has not completed discrete mathematics.
- (c) Every student has completed at least one of the two courses.
- (d) There is a student who has completed exactly one of the two courses.

For each part, write the formal expression clearly.

Problem 3:

(1+1.5+1.5+1=5)

“For every integer n , if n is odd and greater than 1, then there exists a prime number p such that p is less than or equal to n and p divides n .”

Let the domain be the set of all integers, $\mathbb{Z}$. Let the predicates be $O(x)$: “ x is odd” and $P(x)$: “ x is prime.”

- (a) Translate the italicized statement into a formal logical expression using quantifiers ($\forall, \exists$) and the predicates $O(x)$ and $P(x)$.
- (b) Write the negation of your expression from (a). Simplify fully using De Morgan’s laws and quantifier negation rules.
- (c) Translate your final simplified negated expression from (b) back into English.
- (d) A student writes the following fully simplified negation of the italicized statement in English: *“There exists an integer n such that n is odd and greater than 1, and there exists a prime p such that p is not less than or equal to n or p does not divide n .”* Identify the error in the student’s negation and explain why its incorrect.

Problem 4:

(3)

Let $x \in \mathbb{R}$ with $x > -1$. Prove that $(1+x)^n \geq 1+nx$ for all $n \in \mathbb{N}$.

Problem 5:

(1+1+1+1=4)

Prove or disprove the following:

- (a) For all $x \in \mathbb{R}$, if x is rational, then $x^4 + x^2 + 1 > 0$.
- (b) For all $n \in \mathbb{N}$, $n^2 - n + 41$ is prime.
- (c) If p and $p+1$ are both prime and $p > 2$, then p^2 is even.
- (d) For all $p, q \in \mathbb{Q}$ with $p < q$, there exists $r \in \mathbb{Q}$ such that $p < r < q$.

Problem 6:

(5)

Prove or disprove this statement: An 8×8 chessboard cannot be covered by 15 T-tetrominoes and 1 square tetromino. (T-tetromino: ) (Square-tetromino: )

Problem 7:

(5)

Prove that for every integer n , the number $n^7 - n$ is divisible by 42.

Problem 8:

(5)

Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.